

Springer Texts in Statistics

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An Introduction to Statistical Learning

with Applications in R

 Springer

ANALYTICS

Exam 1 Review

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*Examples Inspired By
CAS, SOA, Coaching
Actuaries ©*

Exam #1 Info

- 15 True/False conceptual questions (1×15) + 5 Numerical Free Response questions (5×5 points) = 40 Points
- 75 Minutes
- See Study Guide for Q Distribution – This review won't cover all the topics.

Q1 T/F Practice

- 1. Unsupervised problems are problems without a clear response.
- 2. A qq-plot plots how similar a variable's actual distribution is to a theoretical distribution, using quantiles.
- 3. Categorical variables can only take on one or more of a limited number of different values.

Q2 T/F Practice

- 4. The correlation coefficient represents the percent of variability in the response about its mean explainable by the linear relation between the response and predictor.
- 5. In a simple linear regression model, the $x\%$ confidence interval will coincide with the $x\%$ prediction interval only if irreducible error is 0.
- 6. Classification problems are problems that include a qualitative predictor.

Q3 T/F Practice

- 7. For basic linear regression, if ordinary least squares is used to fit the parameters, the variance of the irreducible error is estimated by the model's mean-squared error.
- 8. As a model increases in complexity, the variance of the response will increase.
- 9. In a simple linear regression, both the confidence interval and prediction interval should be narrowest at the sample mean of X .

Q4 T/F Practice

- 10. An investment analyst interested in finding out if a meme stock is undervalued or overvalued based on the financial characteristics is using modeling for inference.
- 11. When a statistical model overfits the data, it will tend to have low training mean-squared error.
- 12. The error term ε in a supervised learning model is assumed to have a mean of zero.

Q5 T/F Practice

- 13. In simple linear regression with one predictor, the regression sum of squares exceeds residual sum of squares when the coefficient of determination exceeds 0.5.
- 14. Typically, the training data error rate is lower than the test error rate.
- 15. Regression sum of squares represents the amount of variability in the response variable.

Q6

- A K-nearest neighbors (KNN) model is used to classify whether an individual prefers Marvel or Star Wars based on two predictors X_1 (TV Watched), and X_2 (Books Read).
- Make predictions for the following 2 scenarios:
- TV Watched = 1, Books Read = 9, $K = 3$
- TV Watched = 6, Books Read = 4, $K = 5$

TV Watched	Books Read	Prefers
2	1	Marvel
2	7	Star Wars
3	6	Marvel
4	9	Marvel
5	4	Star Wars
6	1	Star Wars
6	6	Star Wars
7	4	Marvel

Q7 (SOA)

- You fit a regression model with one predictor plus an intercept to five pairs of observations.
- The residuals for the first four observations are 0.4, -0.3, 0.0, -0.6, and the unbiased sample variance of the dependent variable y is 1.4.
- Calculate the coefficient of determination.

Q8

- A dataset has 4 observations and is fit to a simple linear regression model for predicting test score using # of coffees drank beforehand.
- If we test $H_0: \beta_1 = 4$ vs. $H_a: \beta_1 \neq 4$, calculate the smallest α from the Tables for which we would reject the null hypothesis.

Test Score	1	9	5	1
Coffees	1	2	3	4

Q9

- You use a simple linear regression model on 3 observations.
- $(y_3 - \bar{y})^2 = \sum_{i=1}^3 e_i^2 + 75$
- $y_3 > \bar{y}$
- Find all missing values in the table.

i	x_i	y_i	$\hat{y}_i$	e_i
1	1	9	10	e_1
2	x_2	22	$\hat{y}_2$	e_2
3	5	y_3	$\hat{y}_3$	-1

Q10

- You fit a linear regression model $Y = \beta_0 + \beta_1 x + \varepsilon$ to explore the relationship of salary (Y) with exams passed (X) on 18 observations.
- Total sum of squares is 862.
- $\bar{x} = 3, \bar{y} = 62$
- $b_1 = 5.2$
- $\sum_i (x_i - \bar{x})^2 = 28$
- Calculate the 90% confidence interval for b_1 .